The transverse rigidity of the particle arrangement: hypotheses and tests

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Abstract

The equilibrium runs (`campaigns.tex`) show transverse sound: the shear mode oscillates with $c_t = 0.022$ instead of diffusing. This note separates what is measured from what is conjectured, states the conjectures with the tests that can falsify them (all in the campaign `rigidity`, now running), and lists ways to remove the rigidity, to be discussed before any is implemented.

I. WHAT IS MEASURED

All at $h/a = 2$, Wendland C2 kernel, $\delta v = 0.043$ ($s_a = 541$), $\eta/s = 0.2$ ($\nu = \eta/w = 2.2 \times 10^{-3}$).

1. $C_T(k,\tau)$ is fitted, within its errors, by the velocity correlator of a damped oscillator,

$$C_T(k,\tau) = A\, e^{-\gamma\tau}\Big[\cos\omega\tau - \frac{\gamma}{\omega}\sin\omega\tau\Big], \qquad \omega^2 + \gamma^2 = c_t^2 k^2, \tag{1}$$

   with $c_t = 0.022$ for $kh = 0.13$–0.74, in boxes 1.2 and 2.4, from two starts and two versions of the code. The fit recovers c_t and γ from synthetic data with known values, and returns $c_t = 0$ for a synthetic fluid.

2. $2\gamma/\nu k^2 = 1.1$–1.2 at $kh \geq 0.5$, rising to ≈ 2 at $kh = 0.13$.

3. The local disorder (rms change of nearest-neighbour distances) saturates at $0.3a$: the particles do not exchange neighbours.

4. Ideal SPH alone (no viscous step), box 0.6: from the lattice with thermal momenta the kinetic energy grows from 0.9 to 6.8 times its equilibrium value by $t = 20$, still rising, while H is conserved to 10^{-7}.

5. Ideal SPH alone from random positions (smoothed entropy made uniform): the kinetic energy reaches 420 times its equilibrium value by $t = 5$. The full scheme from random positions heats by 64% in H by $t = 10$.

II. HYPOTHESIS A: A BROKEN RELABELING SYMMETRY GIVES A SHEAR MODULUS

a. The continuum. The energy of a perfect fluid depends on the density and entropy fields only. A volume-preserving relabeling of the fluid elements leaves the fields, and so the energy, unchanged: a transverse displacement $\boldsymbol{u}$ with $\nabla \cdot \boldsymbol{u} = 0$ costs no energy, and there is no shear modulus.

b. SPH. The smoothed entropy at particle a is $S_a = \sum_b s_b W(\boldsymbol{x}_a - \boldsymbol{x}_b)$. On a lattice with equal $s_b = sV$,

$$S_a = s \sum_{\boldsymbol{G}} \tilde{W}(\boldsymbol{G})\, e^{i\boldsymbol{G}\cdot\boldsymbol{x}_a}, \tag{2}$$

a sum over the reciprocal lattice. The term $\boldsymbol{G} = 0$ is the exact density; the others, set by $\tilde{W}$ at $|\boldsymbol{G}| \geq b = 4\pi/(\sqrt{3}a)$, are the lattice error. A shear at fixed area changes the $|\boldsymbol{G}|$, hence the sum, at second order in the strain ϵ: the energy rises by $\frac{1}{2}G\epsilon^2$ per volume with, by this estimate,

$$\frac{G}{p} \sim C\,(bh)^2 \left|\tilde{W}''(bh)\right| \sim C\, m(m+1) \left|\tilde{W}(bh)\right|, \tag{3}$$

where $\tilde{W} \propto (bh)^{-m}$ asymptotically and C is a number of order the count of shortest $\boldsymbol{G}$ (six). This is an order-of-magnitude argument, not a calculation; in particular the forces in SPH depend on ∇W as well, and the stochastic step may add to or remove from G.

c. Predictions. $\tilde{W}$ at the first reciprocal vector:

kernel	$\langle r^2\rangle/h^2$	$\tilde{W}(bh)$, $h/a = 2$	2.5	3
Wendland C2	0.56	1.2×10^{-4}	3.7×10^{-5}	1.5×10^{-5}
Wendland C4	0.42	3.5×10^{-5}	7.6×10^{-6}	2.2×10^{-6}
Wendland C6	0.34	1.6×10^{-5}	2.3×10^{-6}	5.0×10^{-7}

If $G \propto \tilde{W}(bh)$, then $c_t = (G/w)^{1/2} \propto \tilde{W}(bh)^{1/2}$, and from $c_t = 0.022$ for C2 at $h/a = 2$:

run	c_t if $G \propto \tilde{W}$	c_t if $G \propto \tilde{W}^2$
C2, $h/a = 2.5$	0.012	0.0068
C2, $h/a = 3$	0.0078	0.0028
C4, $h/a = 2$	0.012	0.0064
C4, $h/a = 2.5$	0.0055	0.0014
C6, $h/a = 2$	0.0081	0.0030

d. Falsified if: c_t does not fall with h/a and with the kernel order; or C6 at $h/a = 2$ and C2 at $h/a = 3$ (equal $\tilde{W}(bh)$, different h/a) give different c_t; or the rigidity is absent when the positions are frozen *and* present only with the stochastic step (then it is not a property of H).

e. Second test of the same quantity. The runs record the displacements $\boldsymbol{u}_a$ from the initial sites. For an elastic medium $\langle|\dot{u}_T|^2\rangle = c_t^2 k^2 \langle|u_T|^2\rangle$, and $\boldsymbol{M}_{\boldsymbol{k}} \simeq wV\dot{\boldsymbol{u}}_{\boldsymbol{k}}$, so the statics alone give

$$c_t^2 = \frac{\langle|M_T(\boldsymbol{k})|^2\rangle}{(wV)^2 k^2 \langle|u_T(\boldsymbol{k})|^2\rangle}. \tag{4}$$

In a fluid $\langle|u_T|^2\rangle$ grows linearly in time without bound. Agreement with the fit of eq. (1) is a check that the oscillation is elastic.

III. THE IDEAL HAMILTONIAN'S PHONONS (STATIC TEST)

`examples/ideal_lattice_modes.jl` (branch `rigidity`). On the perfect hexagonal lattice (box 1.2, fixed s_a, zero momenta), displace the particles from their sites by one plane wave,

$$\boldsymbol{u}_a \equiv \boldsymbol{x}_a - \boldsymbol{x}_a^0 = \epsilon\,\hat{\boldsymbol{e}}\cos(\boldsymbol{q}\cdot\boldsymbol{x}_a^0), \qquad \hat{\boldsymbol{e}} \parallel \boldsymbol{q}\ (\mathrm{L}) \text{ or } \perp \boldsymbol{q}\ (\mathrm{T}), \qquad \epsilon = 0.01a, \tag{5}$$

re-smooth, and take the second difference of $U = H(\{\boldsymbol{x}\})$:

$$\Delta U = \tfrac{1}{2}\big[U(\epsilon) + U(-\epsilon) - 2U(0)\big] = \tfrac{1}{2}\epsilon^2\,\frac{N}{2}\,K(\boldsymbol{q}), \qquad \omega^2(\boldsymbol{q}) = \frac{K(\boldsymbol{q})}{wV}. \tag{6}$$

This is the phonon spectrum of the ideal step, from the energy alone: no time stepping, no noise. A fluid has $\omega_L = c_s q$ and $\omega_T = 0$. Results ($c_t = \omega_T/q$ at $qh = 0.26$–0.37; it is constant there):

kernel	$h/a = 2$	2.5	3
C2	$c_t = 0.034$	0.031	0.022
C4	$\omega_T^2 < 0$ (unstable)	unstable	unstable
C6	0.017	0.0066	0.0026

In every case $\omega_L/c_s q = 1.000 \pm 0.003$ at $qh \le 0.37$: the ideal step has the right sound speed.

1. *The rigidity is a property of H.* C2 at $h/a = 2$ has a linear transverse branch with $c_t = 0.034$ on the perfect lattice; the runs give 0.022, the same effect softened by the thermal disorder.

2. *The simple estimate of Sect. II fails quantitatively.* For C2, c_t falls by only 1.6 from $h/a = 2$ to 3 while $\tilde{W}(bh)$ falls by 8; for C4 the sign of the shear modulus is *negative*: the hexagonal lattice is unstable to shear. The shear modulus is a lattice sum whose terms change sign, not a single $\tilde{W}(bh)$. C6 falls fast, a factor 6.5 from $h/a = 2$ to 3, close to $\tilde{W}(bh)^{1/2}$.

3. *Zero modes.* For C2 at $h/a = 2$ (all 648 shells up to $qh = 7.5$), the only vanishing frequencies are at the zone boundary, $qh = 7.255$ ($q = b/2$, neighbouring rows moving in opposite directions): $\omega_L = 0$ and $\omega_T^2 \approx -10^{-6}$. These modes do not change H to second order, so nothing restores them. They are a candidate for the growth of the kinetic energy in ideal SPH (fact 4: energy flowing into neutral modes through the anharmonic terms) and for what the $\lambda = 1$ noise drives into the melt.

a. Larger h/a. The same test for $h/a = 2$–8, with the amplitude fixed by the strain, $q\epsilon = 10^{-2}$ (repeated at 3×10^{-2} with the same results, so these are not round-off). Signed c_t at $qh \simeq 0.26$ (negative: $\omega_T^2 < 0$, unstable to shear):

kernel	$h/a = 2$	3	4	5	6	8
C2	$+0.035$	$+0.022$	-0.011	-0.0034	$+0.0069$	-0.0027
C4	-0.0054	-0.0029	$+0.0029$	$+0.0008$	-0.0008	$+0.0003$
C6	$+0.017$	$+0.0026$	-0.0003	$+0.0002$	$+0.0002$	$< 10^{-4}$

The shear modulus goes to zero as $h/a \to \infty$, as the continuum limit requires, oscillating in sign with a decaying envelope: slowly for C2 (about $(h/a)^{-1.7}$), faster for C4, and fastest for C6, which is below 3×10^{-3} from $h/a = 3$ on. The requirement $c_t \lesssim \pi\nu/L \approx 3 \times 10^{-3}$ for $L = 2.4$ is met on the perfect lattice by C6 at $h/a \geq 3$, C4 at $h/a \geq 4$, and C2 only near $h/a \approx 8$. Whether the thermally disordered state follows the perfect lattice is for the runs to show.

b. Disordered arrangements. The same test on jittered lattices (each site displaced by up to $\pm ja/2$, then the s_a rescaled so the smoothed entropy is uniform; three configurations each; C2, $h/a = 2$). A disordered arrangement is not at an energy minimum, so the symmetric second difference of an imposed plane wave gives the *affine* (Born) modulus only; letting the particles relax non-affinely lowers it, so these are upper bounds.

	lattice	$j = 0.1$	0.3	1	random
affine c_t ($qh \simeq 0.26$)	0.034	0.034	0.034	0.049	0.086
$\omega_L/c_s q$	1.002	1.002	1.001	1.003	1.01

Disorder does not restore the relabeling symmetry: the affine modulus is unchanged for small disorder and *grows* for large disorder, to $c_t = 0.086$ for random positions. The thermal state of the runs (nearest-neighbour disorder $0.3a$) has the dynamic $c_t = 0.022$ against the affine 0.034: the difference is the non-affine relaxation (and anharmonicity), about 35% in c_t. So randomizing the particles is not a cure; it makes the arrangement stiffer.

Consequences: (i) G can be positive or negative depending on kernel and h/a, so it can be made small, perhaps zero, on the perfect lattice by the choice of kernel and h/a; but the relevant state is the thermally disordered one, where c_t is smaller (0.022 against 0.034). (ii) The C4 runs now going start on a lattice that is unstable to shear. (iii) Any revision of the ideal step, including $H = \sum_a e_a$, should be checked first with this static test: $\omega_L = c_s q$, $\omega_T \to 0$, and no zero or negative modes in the zone.

IV. SUMMARY: AN EXPLICITLY BROKEN SYMMETRY AND TWO PARAMETERS

The kernel-sum energy breaks the relabeling symmetry of the fluid *explicitly*, by an amount $\epsilon(h/a, \text{kernel})$ that vanishes in the continuum limit $h/a \to \infty$ (the static modulus $G \sim p\,\epsilon$ of Sect. III, oscillating in sign). Given ϵ, whether the particles form a solid or a liquid is a second question, as for molecules: the ratio of the depth of the arrangement landscape to the temperature,

$$\Gamma \sim s_a\,\epsilon. \tag{7}$$

For $\Gamma \gg 1$ (now: $s_a = 541$) the particles are caged, the modulus is static, and the shear mode is transverse sound. For $\Gamma \lesssim 1$ they would exchange neighbours, the modulus would relax in a Maxwell time τ_M, and the broken symmetry would only add a viscosity $G\tau_M$. s_a is set by the physics; ϵ is set by the method. A smaller ϵ (kernel, h/a) both shrinks G and moves toward $\Gamma \lesssim 1$; volumes that partition space ($H = \sum_a e_a$) remove the explicit breaking for homogeneous strain altogether.

V. HYPOTHESIS B: THE ARRANGEMENT CARRIES ENERGY

Facts 4 and 5 show that at fixed s_a the SPH energy depends on the arrangement, by a lot for a random one ($\sim 400T$ per particle released) and by a little for the lattice ($\sim 6T$ per particle by $t = 20$, still growing). The conjecture is that at temperature T the equilibrium of the positions lies near a minimum of this energy, so the particles form a solid whatever the start, and a random arrangement is not an equilibrium state.

This is *not* established. Fact 4 says that the hexagonal lattice is not itself the minimum (ideal SPH moves away from it, releasing energy), so the relevant minimum is some nearby disordered arrangement, and why the full scheme stays at a disorder of $0.3a$ is not explained. The ideal-SPH runs at $h/a = 2$, 2.5, 3 and for C4, C6 (to $t = 50$) show whether the kinetic energy saturates, and how the released energy scales with $\tilde{W}(bh)$. If it scales like G, A and B are the same effect.

VI. WAYS OUT, IF A HOLDS

a. Smaller lattice error. A fluid needs $c_t k \ll \nu k^2/2$ at the smallest k of interest,

$$c_t \ll \frac{\nu k_{\min}}{2} = \frac{\pi \nu}{L}, \tag{8}$$

about 3×10^{-3} for $L = 2.4$: a factor 7 in c_t, 50 in G. Whether a kernel and h/a reach this at acceptable cost is what the runs measure.

b. Noise in $d\boldsymbol{x}/dt$. Particles that exchange neighbours relax shear stress in a Maxwell time $\tau_M \sim a^2/D_x$ (diffusion constant D_x of the positions relative to the fluid), turning the solid into a fluid with an added viscosity $G\tau_M$. This is small compared with η when

$$D_x \gg \frac{c_t^2 a^2}{\nu} \approx 0.06\,\nu. \tag{9}$$

Three versions, with their expected outcomes:

1. *Displacements alone* (s_a, $\boldsymbol{m}_a$ fixed): each move raises the arrangement energy (Hypothesis B), which ends up as heat; expected to heat strongly.

2. *Displacements with a Metropolis test on e^S*: samples the true equilibrium of the positions; if B holds that is the solid, and G is unchanged.

3. *Relabeling*: displace a particle and refit s_a, $\boldsymbol{m}_a$, e_a so that the smoothed fields are unchanged, as molecular self-diffusion leaves the hydrodynamic fields unchanged. The energy change is then only the error of the refit, which a Metropolis test can absorb. This restores the relabeling symmetry to the accuracy of the refit. It needs a local linear solve per move that conserves E and $\boldsymbol{M}$, and a check that it does not spoil the detailed balance of the stochastic step.

A quick test of each at $L = 0.6$ (heating and c_t against D_x) would settle 1 and 2.

c. A different Hamiltonian: $H = \sum_a e_a$ with a partition of space (owner's suggestion). Write the energy as the sum of the particle energies,

$$H = \sum_a e_a(s_a, \boldsymbol{m}_a, V_a), \qquad \sum_a V_a = A \text{ exactly}, \tag{10}$$

with the volumes $V_a(\{\boldsymbol{x}\})$ a partition of the area A: Voronoi cells, or the partition of unity $\psi_a(\boldsymbol{x}) = W(\boldsymbol{x} - \boldsymbol{x}_a)/\sum_b W(\boldsymbol{x} - \boldsymbol{x}_b)$, $V_a = \int \psi_a \, d^2x$, of the meshless finite-mass methods. Then:

1. *No shear modulus for homogeneous strain (exact).* An area-preserving affine map takes a Bravais lattice into another Bravais lattice, on which all cells are equivalent, so every V_a stays A/N and H is unchanged: $G = 0$. The present kernel sum $S_a = \sum_b s_b W(\boldsymbol{x}_a - \boldsymbol{x}_b)$ has no such property; its change under shear is the lattice error of Sect. II.

2. *Conjecture:* non-affine motions change the V_a at second order in the strain gradients, so the effective modulus vanishes as $(ka)^2$ at long wavelength, which is what a fluid needs. Test: $c_t(k)$ in a run with eq. (10); it should fall linearly with k.

3. *One energy.* The stochastic step conserves $\sum_a e_a$ exactly; with eq. (10) the Hamiltonian step conserves the same quantity, and the mismatch between the smoothed H and $\sum_a e_a$, a candidate source of the slow heating, is gone by construction.

Two caveats. (a) On a *disordered* arrangement an affine shear does change the individual Voronoi volumes (cell boundaries are perpendicular bisectors, which a shear does not preserve); with equal pressures the first-order change $p \sum_a \delta V_a$ vanishes, but the second-order one, $\sum_a (K/2V)\, \delta V_a^2$, gives an affine modulus $\propto$ (cell-shape disorder). It should relax away: N volume constraints on $2N$ coordinates leave an N-dimensional family of motions at fixed V_a, along which the particles can absorb the shear at no cost, as volume-preserving flows cost nothing in a fluid (an argument, to be checked with the static test after relaxation). (b) That family also contains lattice-scale zero modes (cells degenerating into long thin shapes), which nothing in H restores; moving-mesh codes add a mild regularization (steering each generator toward its cell's centroid). The static test should count them.

Costs: the particles move with their own velocities, $\dot{\boldsymbol{x}}_a = \partial H/\partial \boldsymbol{m}_a$, not smoothed ones, so the lattice-scale modes are less filtered; the forces need $\partial V_b/\partial \boldsymbol{x}_a$ (Voronoi geometry or the partition-of-unity integrals). And with $e_a(s_a, V_a)$ each particle is its own thermodynamic system: the pressure and the equilibrium distribution $e^{\sum_a s_a}$ live at the scale a, and how the coarse-grained pressure and entropy follow from them has to be worked out (the factor 9/16 in the energy variance of `eqstart.tex` is an instance).

d. Not proposed. Lowering s_a to melt the arrangement: the scheme blows up at $\delta v = 0.14$ (`campaigns.tex`).